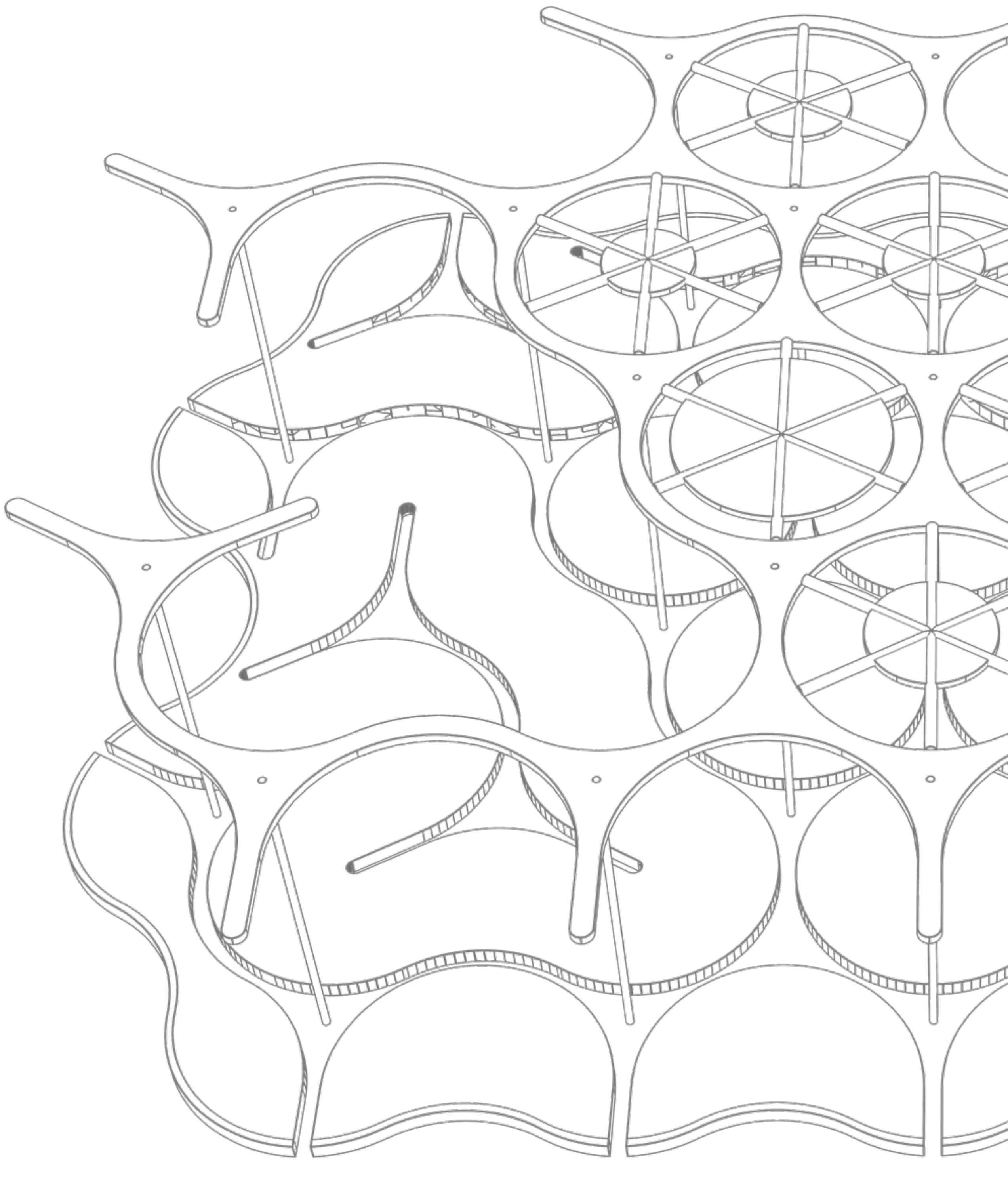
SARA ELIMAM

ARCHITECTURE & BUILDING ENGINEER
SELECTED WORKS



ABOUT ME

An Architectural Designer focused on data-driven sustainability and technical constructability. Merging advanced parametric modeling with rigorous architectural detailing. Bridging the gap between concept and high-performance reality.

EDUCATION

M.Sc. Building & Architectural Engineering
Politechnico di Milano
Score: 108/110 (July 2026)

B.Sc. Architecture (with Honors)
University of Khartoum
Second Class-Division One 7.4/10 (Dec 2021)

EXPERIENCE

Design Engineer
Mo. Siraj Engineering Co. Ltd.
Khartoum, Sudan – Feb 2022 - May 2023
- Conceptualizing, developing and finalizing design projects (private villas and Interior designs) under the lead architects supervision
- Preparing architectural construction drawings (private and public) including interior and exterior visualization of projects
- Specifying finishing materials, preparing BOQs, and coordinating designs with structural engineers.

Adjunct Teaching Assistant,
Faculty of Architecture
Khartoum, Sudan – June 2022 - Dec 2022
- Served as an adjunct Teaching Assistant for Construction Technology courses.
- Supported course delivery by expanding relevant construction technology content and providing real market examples.
- Assisted in preparing and grading assignments, tutoring students, and providing regular feedback to both students and professors on academic progress.

Architectural Intern,
AL BARAJOUB ENGINEERING CO. LTD
Khartoum, Sudan – Sep 2018 - Oct 2018
- Introduced to the environment and workflow of projects in the architectural office
- Assisted resident engineers in 3 sites (finishes, structural concrete casting, measurements take off)

CERTIFICATION & COURSES

Global Youth Convention-3rd, HISA, Dec 2025
Construction LCA Bootcamp, One Click LCA, Nov 2025
Parco Mamaor Eco Villaggio Prosperità Globale, Politechnico Di Milano, Sep 2024
Energy Simulations: Ladybug Tools, Udemy, Oct 2025
Autodesk Revit, Udemy, Dec 2025

SKILLS

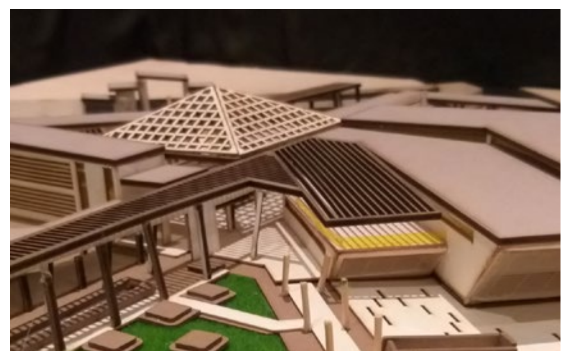
Design & BIM: AutoCAD, Revit Architecture, Navisworks, Rhino + Grasshopper
Analysis & simulations: Climate studio, Ladybug, Ecotect
Visualization: Lumion, 3ds Max + Corona, Photoshop, Illustrator, InDesign
Other: Python Programming Language, Microsoft Office, Manual Sketching & Model Making

LANGUAGES

Arabic, Native
English, IELTS C1 (Dec, 2022)
Italian, CPIA A2 (Jan, 2026)

CONTACT

Milan Area, Italy
Email : sarahalemam.37@gmail.com



CONTENTS

FACADE LAB

Laurea Magistrale Thesis

To check the full project Report on Polimi repositary. scan the QR code

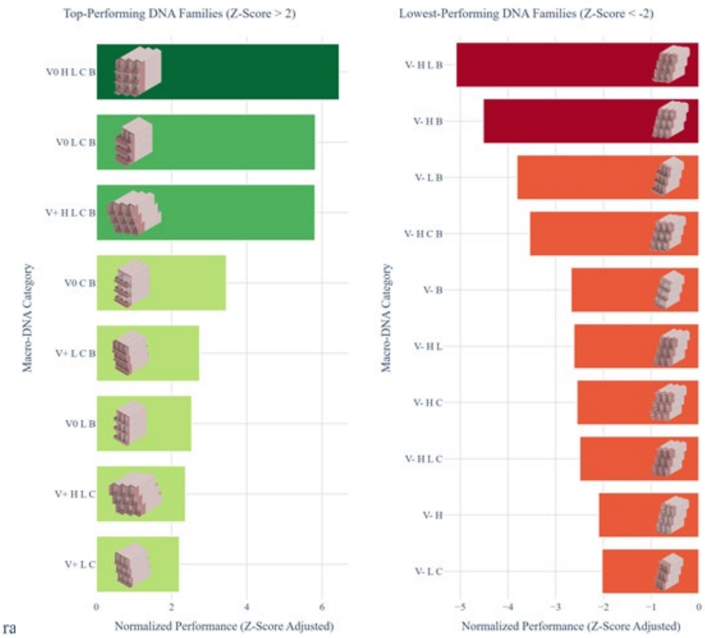
Thesis Collaboration with Hagar Hammad

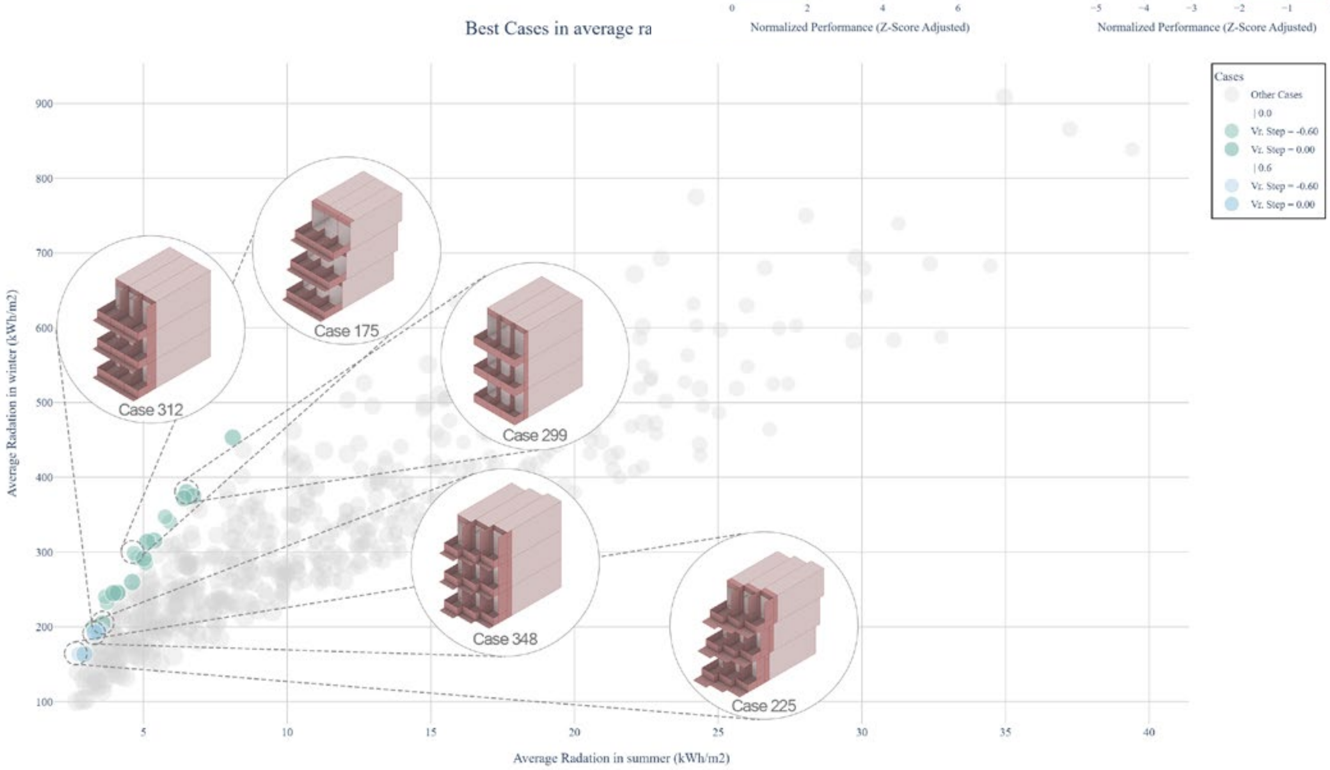
Date and Location:
July 2026, Milan, Italy
Software:
Rhino, Grasshopper (Ladybug/Honeybee), Jupyter Notebook, Python

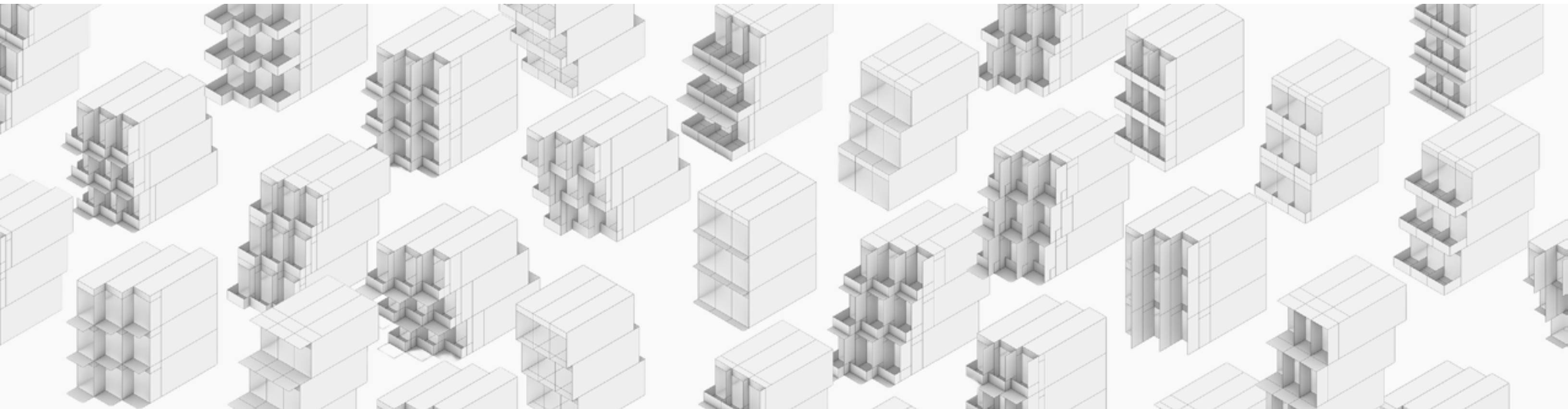
Description:
This thesis transforms façade design from simulation-heavy evaluation into AI-assisted exploration. Over 1,000 morphological variations were simulated to build a performance dataset, then encoded as Digital Elevation Models (DEMs) for a Convolutional Neural Network, achieving $R^2 = 0.955$ in predicting thermal, daylight, and active solar performance. The resulting framework enables near real-time environmental feedback, bringing data-driven sustainability directly into early-stage design.

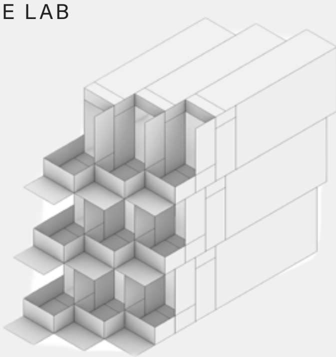
Data Production and analysis

Data points were parametrically generated and analyzed through Pareto frontiers in distinct domains, each featuring contradicting environmental targets. The data was then grouped into specific categories based on architectural morphology. Finally, group Z-scores were applied to rank the data points' overall performance stability across these objectives. The front elevation of each generated case was then used to train the CNN Machine Learning model.



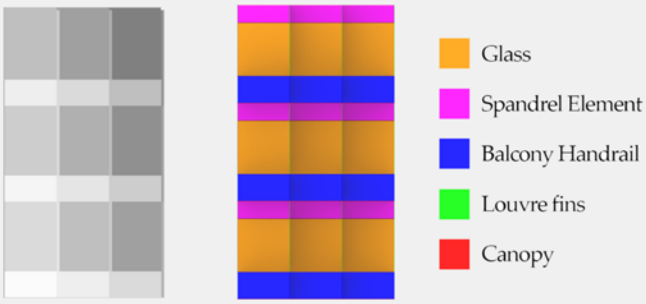






BACK-END DEVELOPMENT

Predictive accuracy $R^2 = 0.95$



Winter Radation
Summer Radiation
sDA
ASE
Active Generation

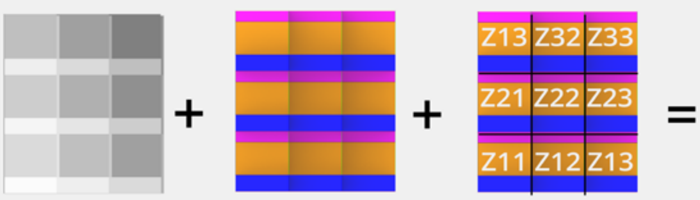
Y = Normalised Performance vector

Dataset split:
70% Training
15% Validation
15% Test
Epochs used is 50
batch size of 16
Best model is selected using **validation loss**

1 Inputs and compression for compatibility

The system processes two types of inputs simultaneously, formatted to a standardized square images. The Digital Elevation Model (DEM) and the semantic color-coded map divided into five separate layers, each representing a distinct architectural feature.

2



Geometric feature vector

DEM Encouder

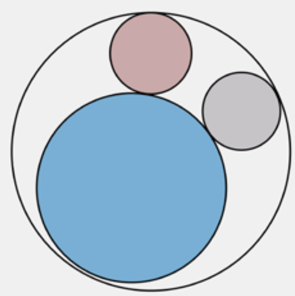
Multichannel Encouder

Zone Pooling

Feature Fusion head

3

4



Encouder, Fusion, and Regression head

The pipeline maps visual inputs (X)—combining geometric (X_dem) and spatial-semantic (X_sem) features—to a 5-performance indicator vector (Y). During training, the targets (Y) are normalized (Y_norm) to balance scales. A regression head processes the fused X representation to predict normalized outputs (Y_pred), which are denormalized during testing to recover physical units. The final model achieved high accuracy, yielding an average R_ of 0.955

Regression Head 5

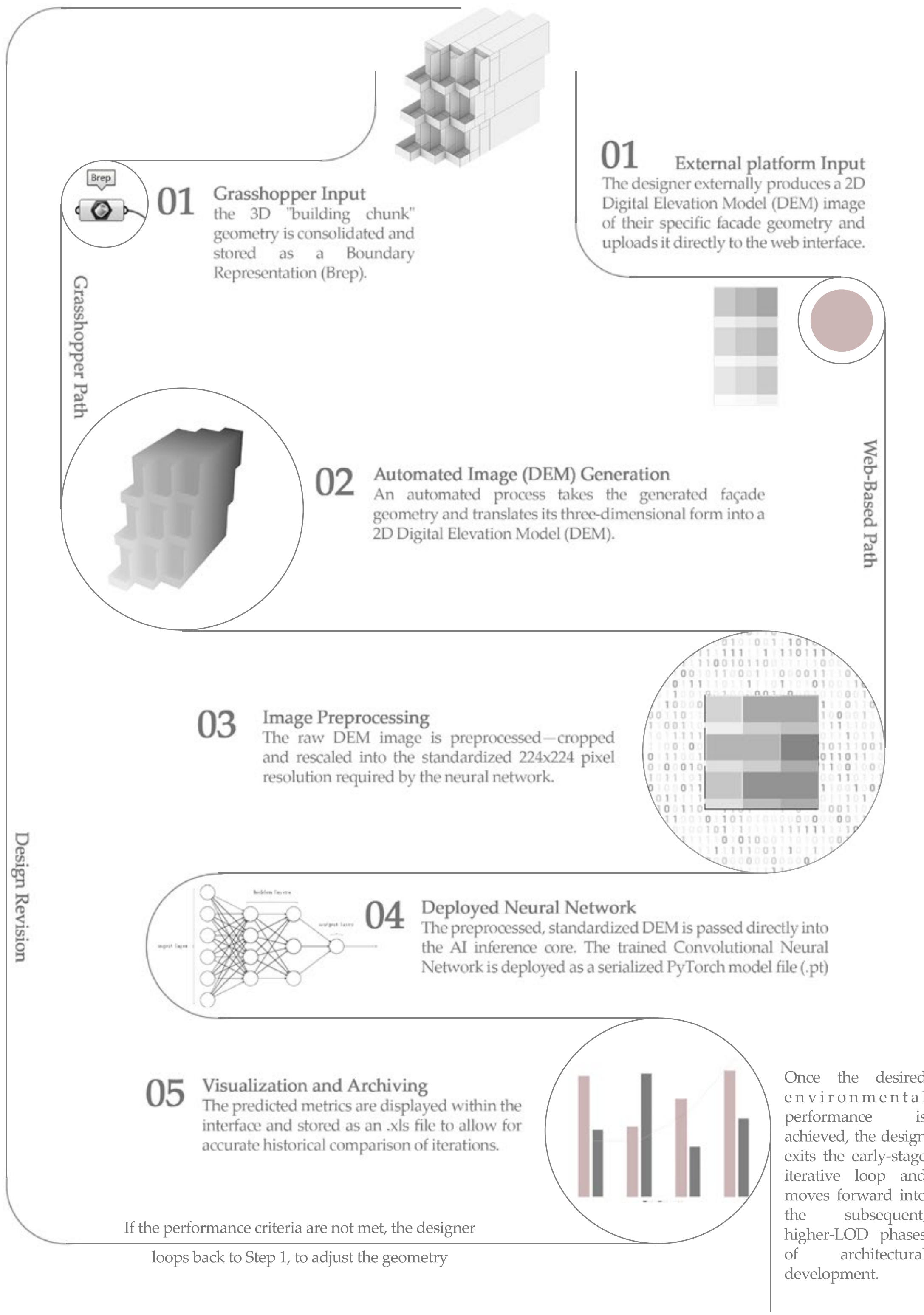
Predections

6

R^2 — the closer to one the better

MAE — the closer to zero the better

FRONT-END DEVELOPMENT



Click or scan the qr code to access the prototype! (restart the space if error time encountered)

The framework deploys via two interfaces:

The framework deploys via two interfaces:
Web Dashboard:
A prototype providing rapid, ML-driven environmental predictions. It translates DEM Images into predicted values

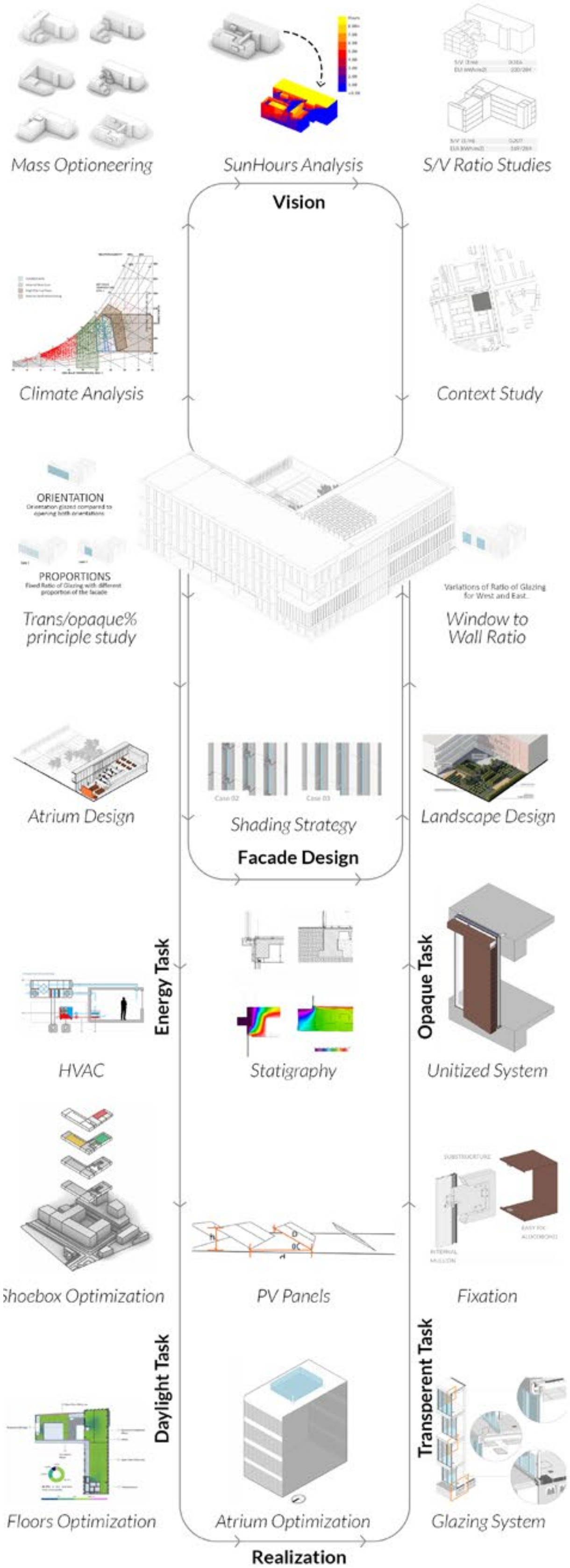
Grasshopper Integration:
A custom native tool that automatically extracts 3D geometry, returning real-time performance feedback directly onto the design canvas.

PARASITE
Mentored by Park Associati

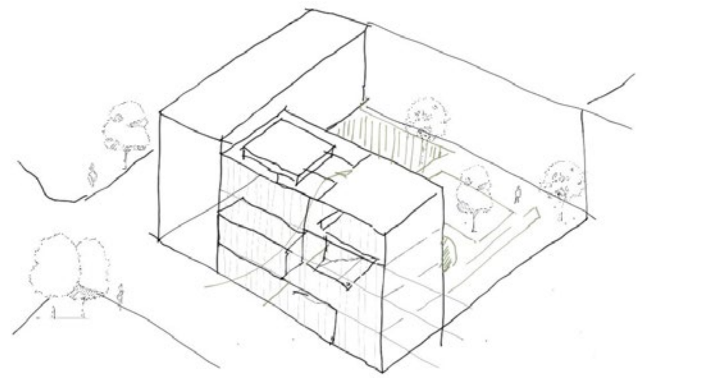
Date and Location:
Jan 2025, Milan, Italy

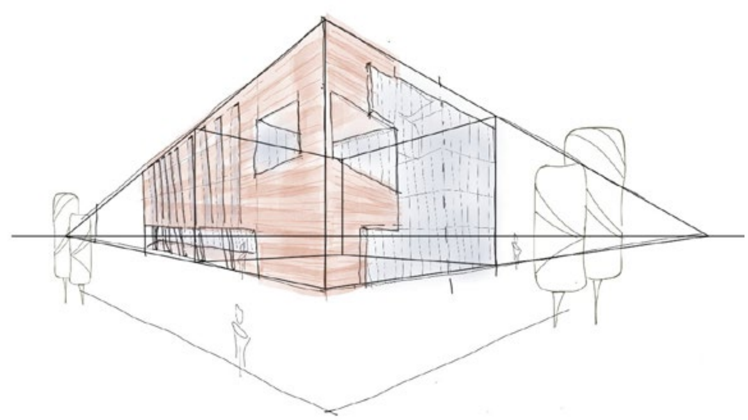
Type:
Architectural Design & SBT Studio
Software:
Rhino, Grasshopper (Ladybug/Honeybee), CAD/BIM

Description:
Mentored by Park Associati, this integrated studio tackles the adaptive reuse of an existing office building. To harmonize with Milan's historic urban fabric, the primary mass is reclad in heavy terracotta. In sharp contrast, the design introduces a lightweight, modern "parasitic" intervention housing an innovative urban farming lab. A highly interactive bioclimatic core acts as the building's "lung," regulating passive ventilation while seamlessly connecting the indoor and outdoor public realms.



Workflow
Integrated Design

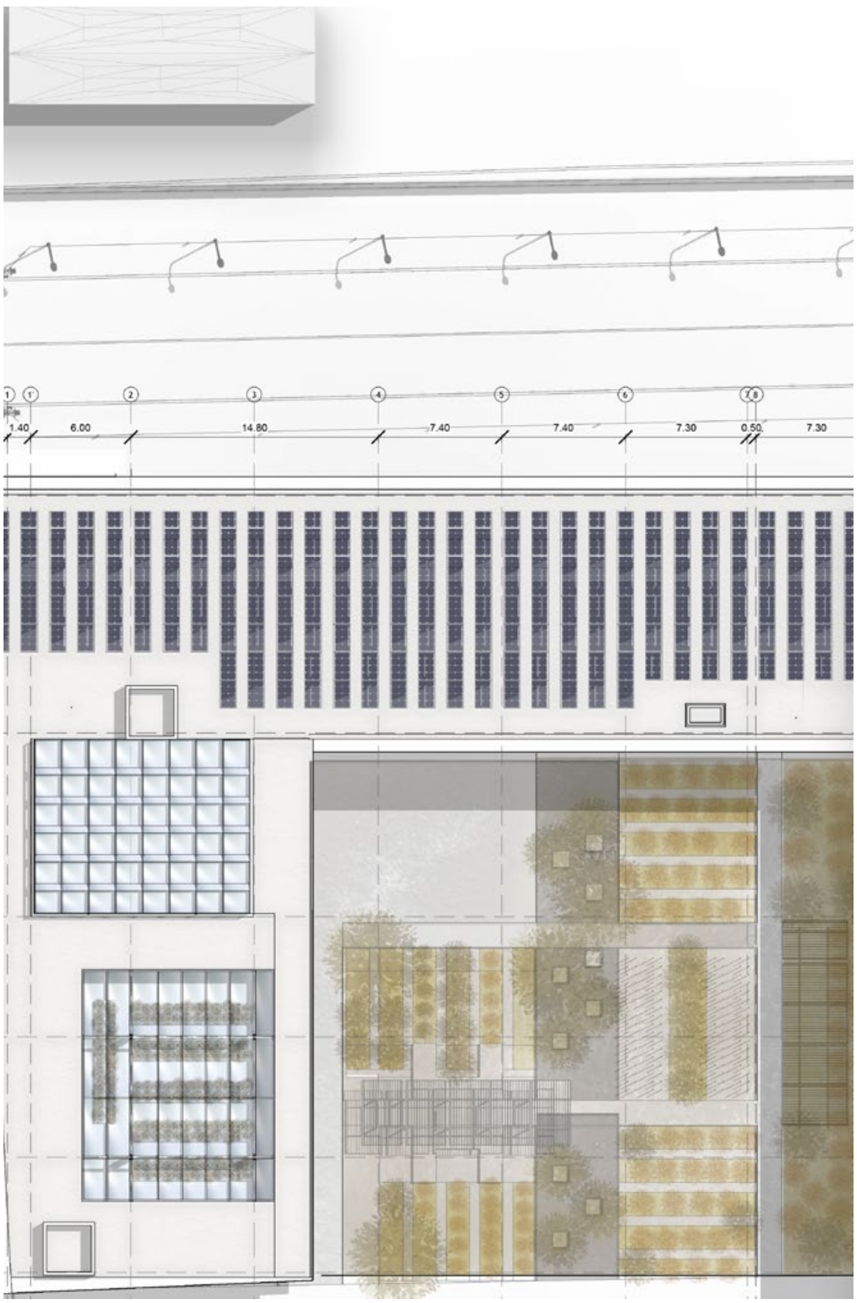


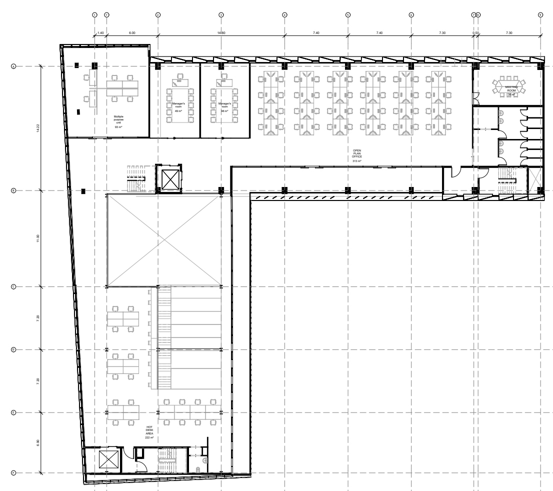


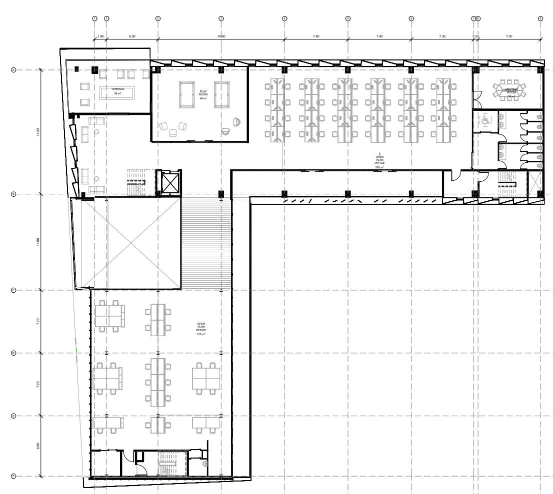
The project was driven by a highly integrated workflow, simultaneously balancing the conceptual geometry of the parasitic form with rigorous technical performance constraints. By continuously adjusting the architectural massing in direct response to structural and energy feedback, the design achieves a flawless fusion of bold aesthetics and environmental engineering.

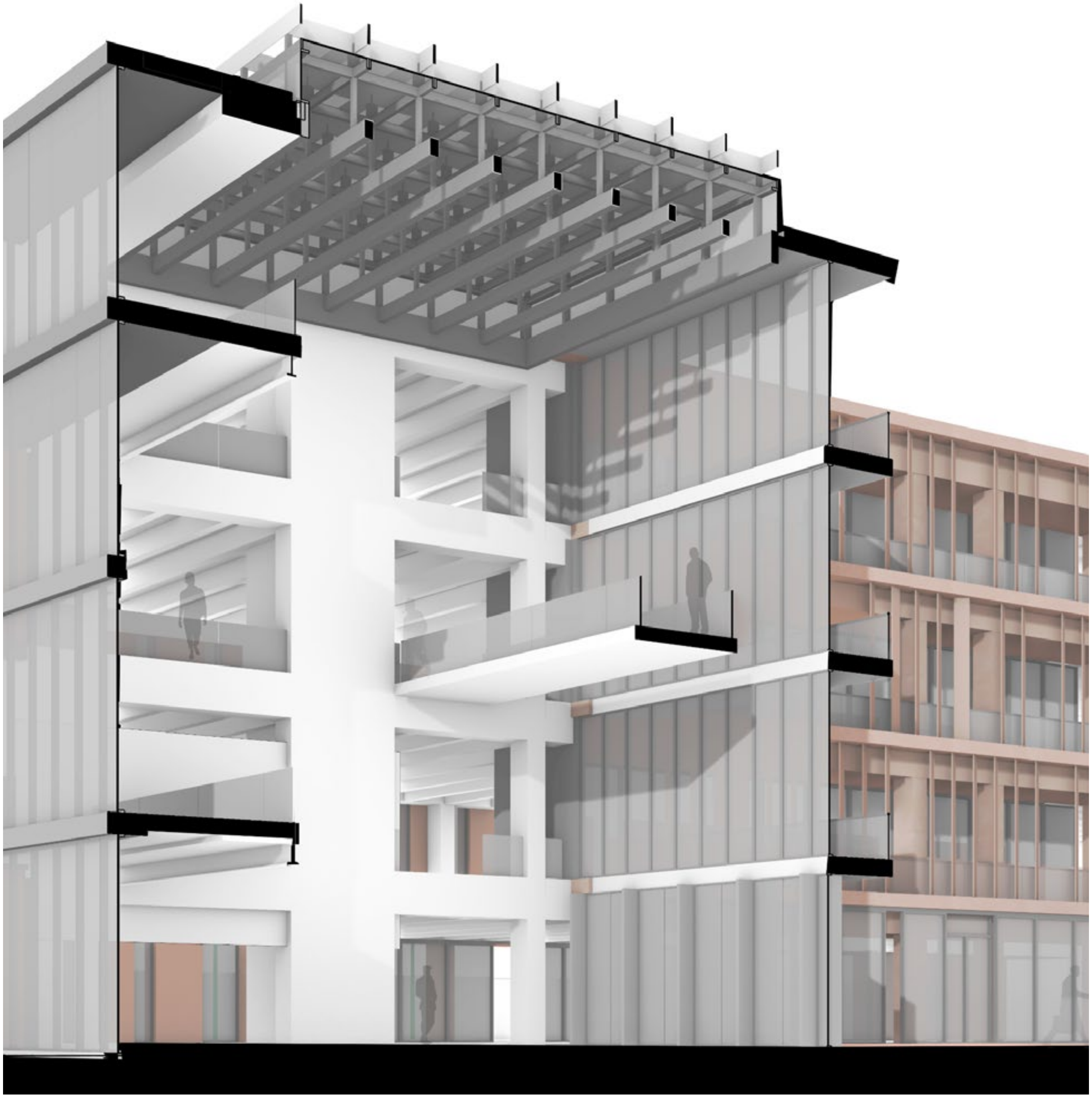
Using a continuous parametric feedback loop, early-stage 3D models were directly linked to environmental simulations. This data-driven approach allowed for real-time testing of the lightweight addition against the historic terracotta mass, ensuring the complex geometry was fully optimized for daylight and energy efficiency before advancing to the technical detailing phase.

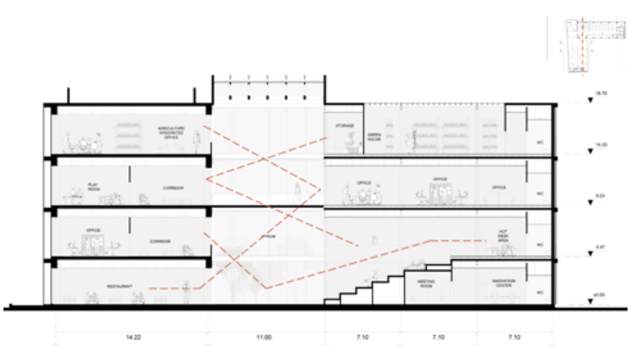
Functionally, the floor plans are strictly zoned to emphasize the architectural contrast. Standardized workspaces are efficiently organized within the original footprint, while the newly introduced volumes are dedicated entirely to the agricultural research and flexible collaborative zones. Extending this programmatic logic outdoors, the masterplan introduces a curated seasonal garden. This responsive landscape adapts to Milan's changing climate, pulling the urban farming ethos into the public realm and providing a continuously evolving natural amenity for the users.



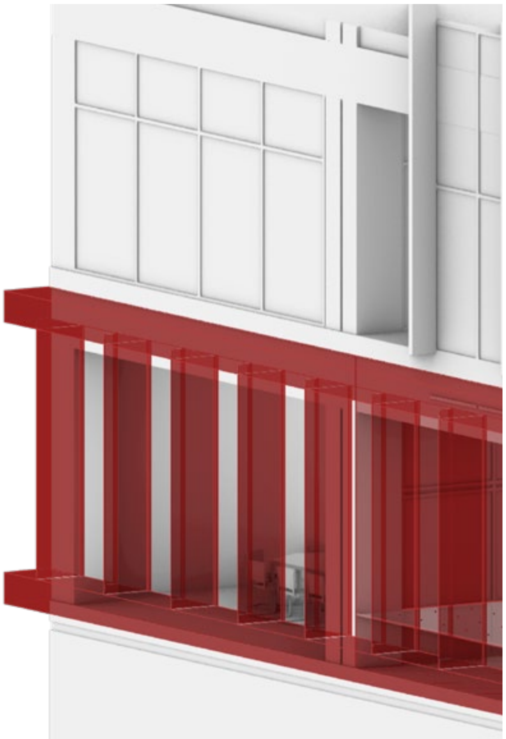


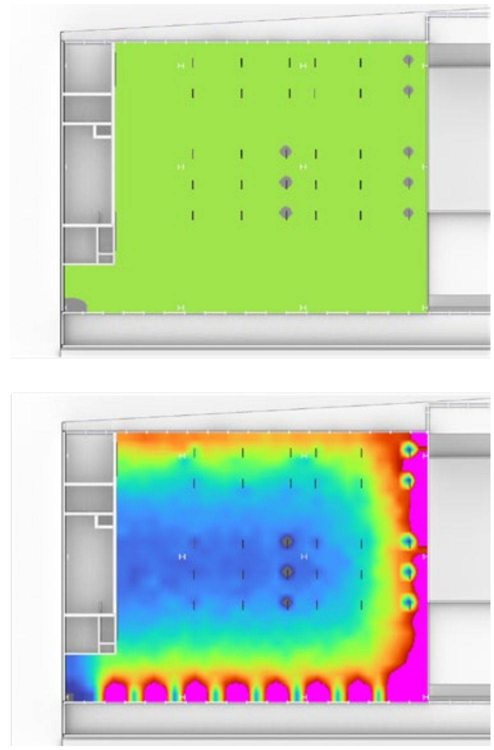






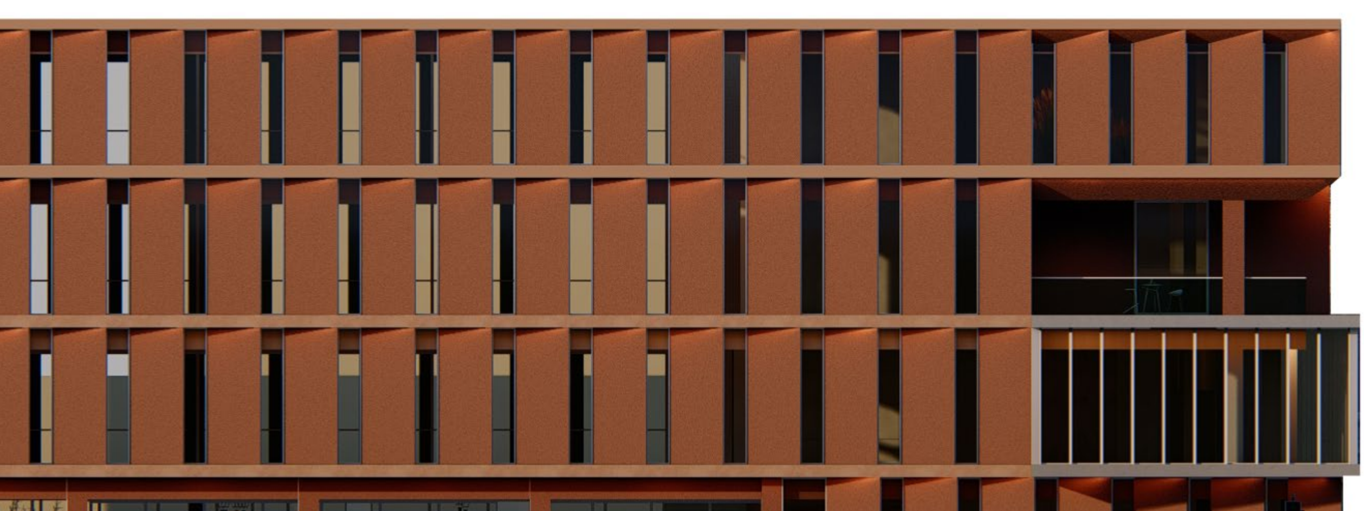
Anchoring this spatial layout is the central core. Beyond its role as the primary social circulation hub, this vertical atrium functions as a thermodynamic engine. By harnessing the stack effect, it actively drives the passive ventilation strategies established in the initial concept. Furthermore, it acts as a visual and physical bridge between the enclosed office floors and the outdoor seasonal gardens, fostering spontaneous interaction while naturally optimizing the internal microclimate.

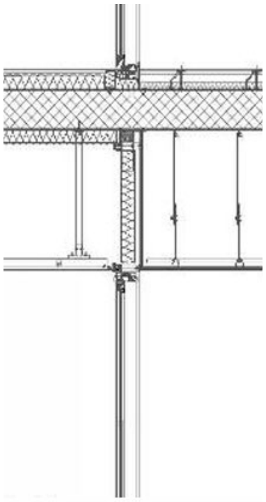


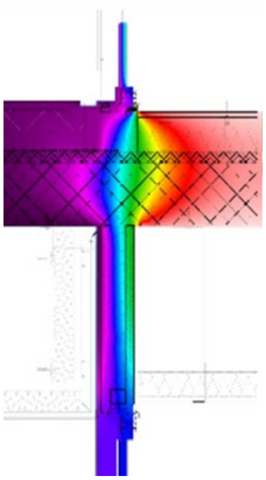


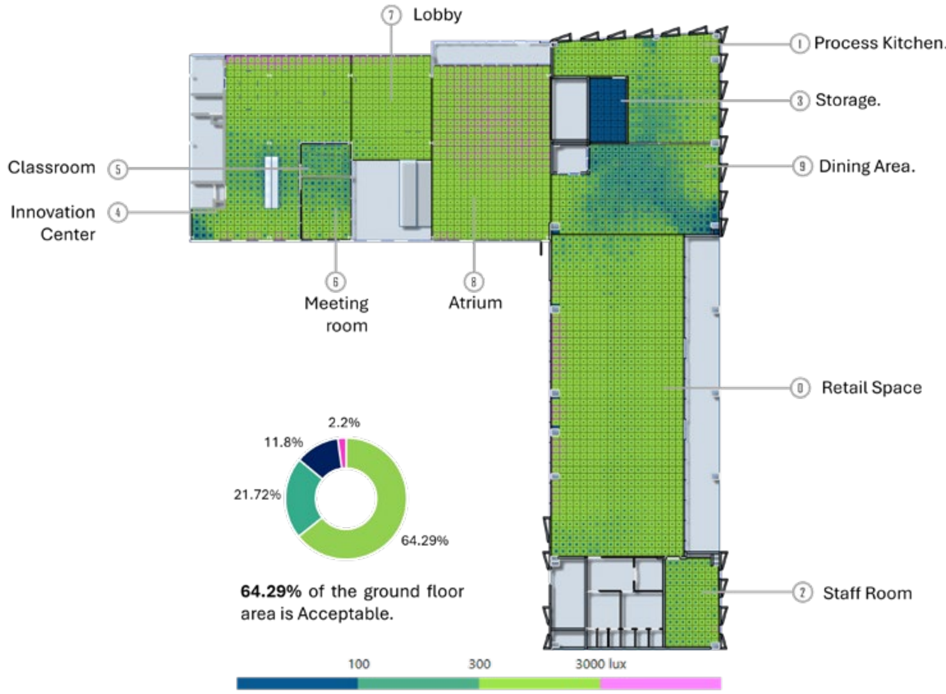
Utilizing Grasshopper, Ladybug, and Honeybee, parametric simulations were conducted to test iterative facade alternatives across all orientations. A standout intervention is the East elevation's custom ventilated curtain wall. This unitized system strategically redirects harsh morning sun toward the north, allowing for precise daylight control and significantly reducing interior cooling loads.

To optimize computational efficiency, the visual comfort analysis employed a multi-scalar workflow. Daylighting strategies were initially tested on localized "shoebox" models across various solar orientations. Once calibrated for optimal Spatial Daylight Autonomy (sDA) and low ASE glare, these baseline solutions were seamlessly scaled and fine-tuned across the entire parasitic geometry.

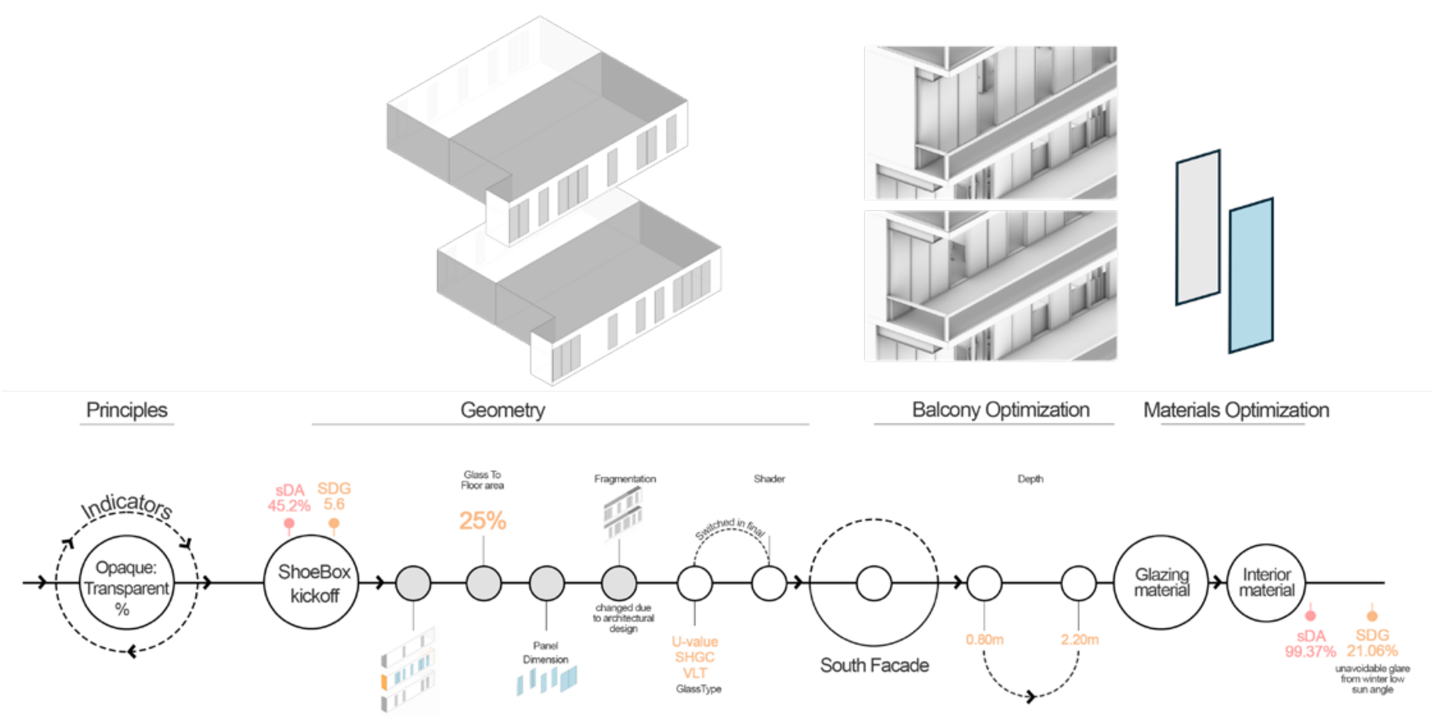




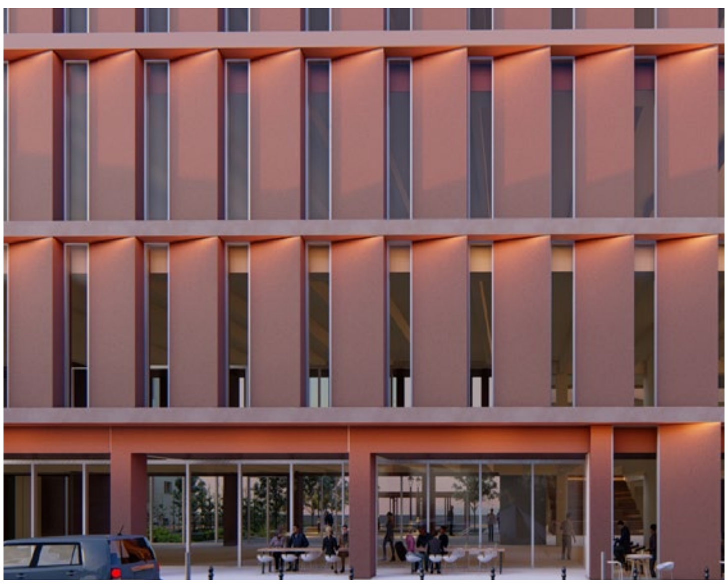


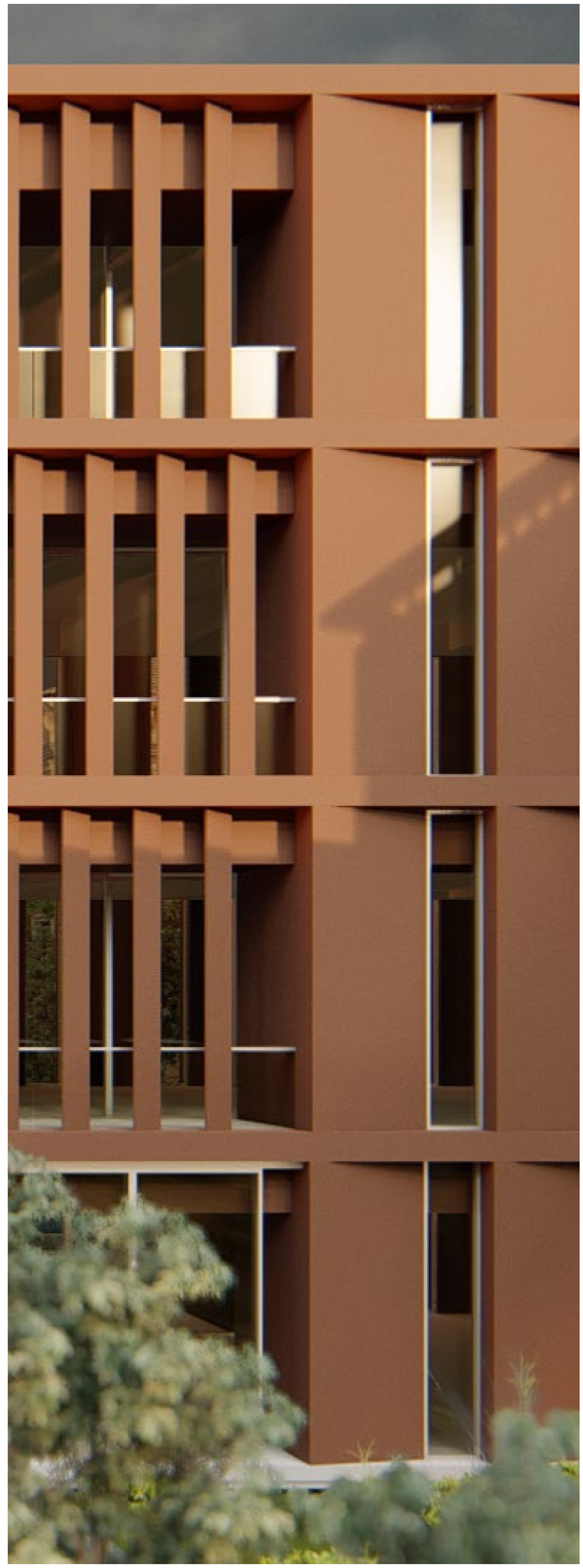


The integrated design successfully achieved an A2 Energy Class rating. The optimized floor plans demonstrate exceptionally high Spatial Daylight Autonomy (sDA) while strictly minimizing Annual Sunlight Exposure (ASE). Furthermore, advanced analyses of critical system connections successfully eliminated thermal bridges, ensuring envelope efficiency and mitigating expensive long-term maintenance.

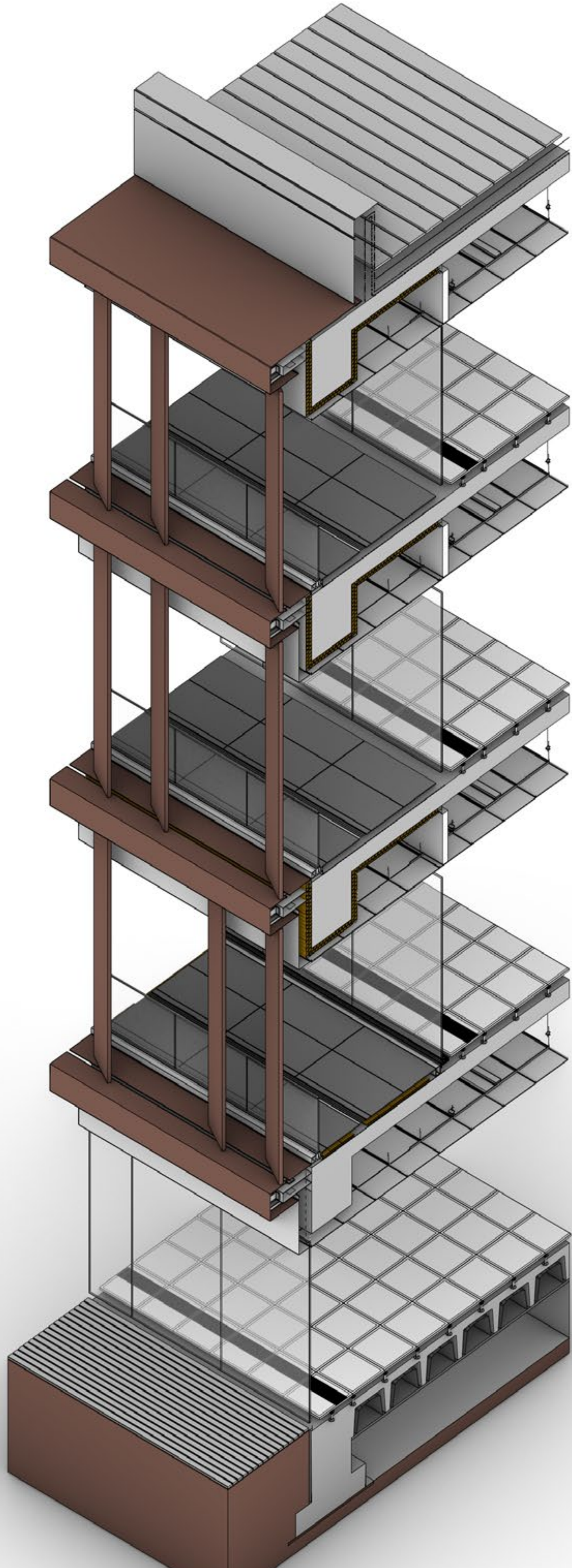


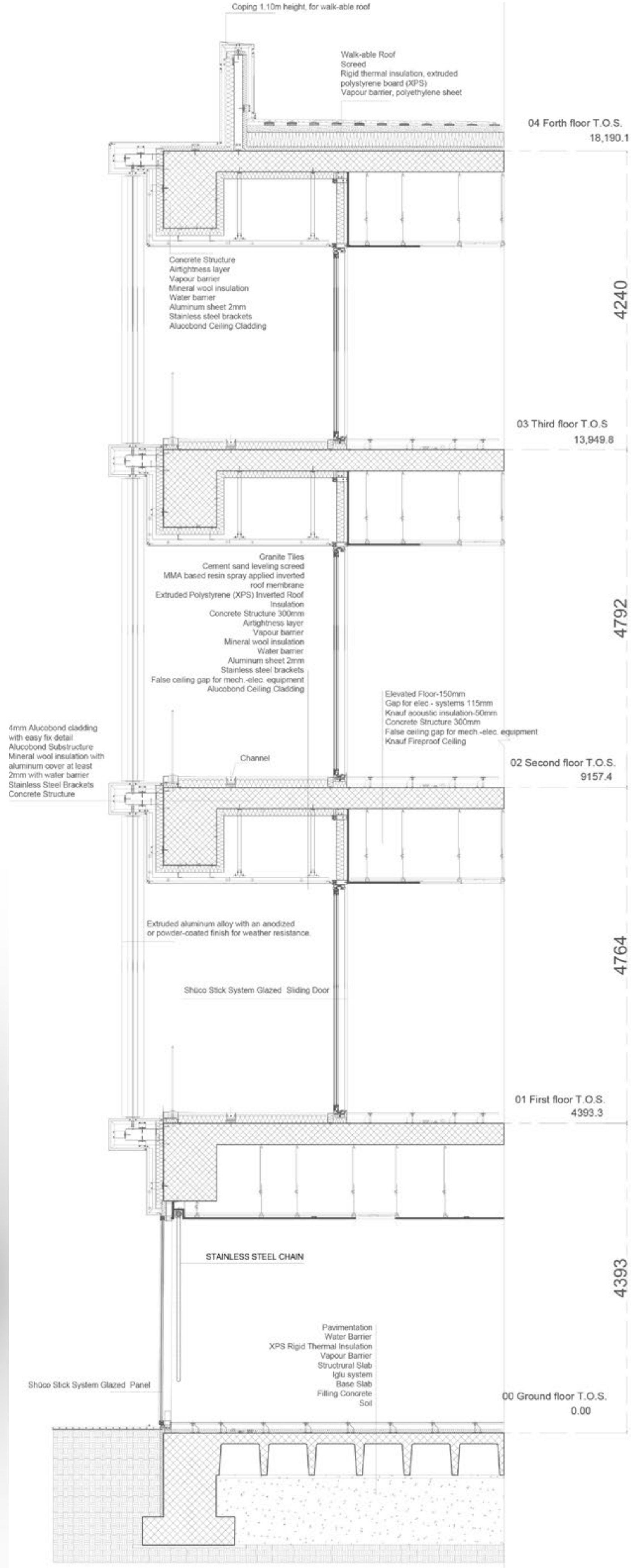
A2
Energy Class





To ensure constructability, the technical sections rigorously resolve the building's envelope by tracking the continuous insulation line from the roof down to the foundation. At the base, an Iglu' ventilated system manages rising damp, while the modular curtain wall above perfectly integrates the transparent glazing and opaque elements. The strict thermal continuity is maintained at the complex junctions between the old mass and the modern lightweight intervention. Because system connections often represent the weakest points in construction, these specific junctions were rigorously tested for thermal bridging, proving the parasitic geometry can be flawlessly executed as a watertight, energy-efficient architectural skin.





Coping 1.10m height, for walk-able roof
Walk-able Roof
Screed
Rigid thermal insulation, extruded polystyrene board (XPS)
Vapour barrier, polyethylene sheet
04 Forth floor T.O.S.
18,190.1
Concrete Structure
Airtightness layer
Vapour barrier
Mineral wool insulation
Water barrier
Aluminum sheet 2mm
Stainless steel brackets
Alucobond Ceiling Cladding
4240
03 Third floor T.O.S
13,949.8
Granite Tiles
Cement sand leveling screed
MMA based resin spray applied inverted roof membrane
Extruded Polystyrene (XPS) Inverted Roof Insulation
Concrete Structure 300mm
Airtightness layer
Vapour barrier
Mineral wool insulation
Water barrier
Aluminum sheet 2mm
Stainless steel brackets
False ceiling gap for mech.-elec. equipment
Alucobond Ceiling Cladding
4792
Elevated Floor-150mm
Gap for elec - systems 115mm
Knauf acoustic insulation-50mm
Concrete Structure 300mm
False ceiling gap for mech.-elec. equipment
Knauf Fireproof Ceiling
4mm Alucobond cladding with easy fix detail
Alucobond Substructure
Mineral wool insulation with aluminum cover at least 2mm with water barrier
Stainless Steel Brackets
Concrete Structure
Channel
02 Second floor T.O.S.
9157.4
Extruded aluminum alloy with an anodized or powder-coated finish for weather resistance.
Shüco Stick System Glazed Sliding Door
4764
01 First floor T.O.S.
4393.3
STAINLESS STEEL CHAIN
4393
Pavimentation
Water Barrier
XPS Rigid Thermal Insulation
Vapour Barrier
Structural Slab
Iglu system
Base Slab
Filling Concrete
Soil
Shüco Stick System Glazed Panel
00 Ground floor T.O.S.
0.00

SAN CRISTFORO GARDEN

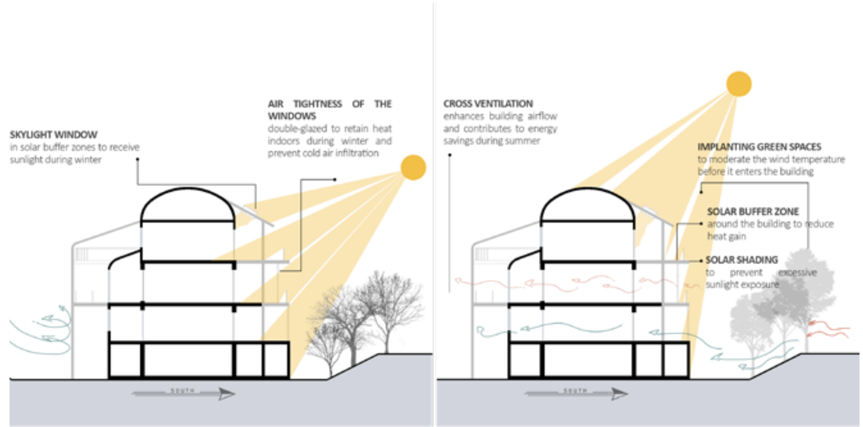
ALDO ROSSI'S EX-TERMINAL SITE

Date and Location:
June 2024, Milan, Italy

Type:
Academic Group Project / Urban Masterplanning

Description:
This urban masterplan regenerates Milan's San Cristoforo district, transforming a fractured, post-industrial railway yard into a vibrant ecological corridor. It successfully reclaims these abandoned urban voids, turning them into active, high-quality public spaces for the surrounding community.

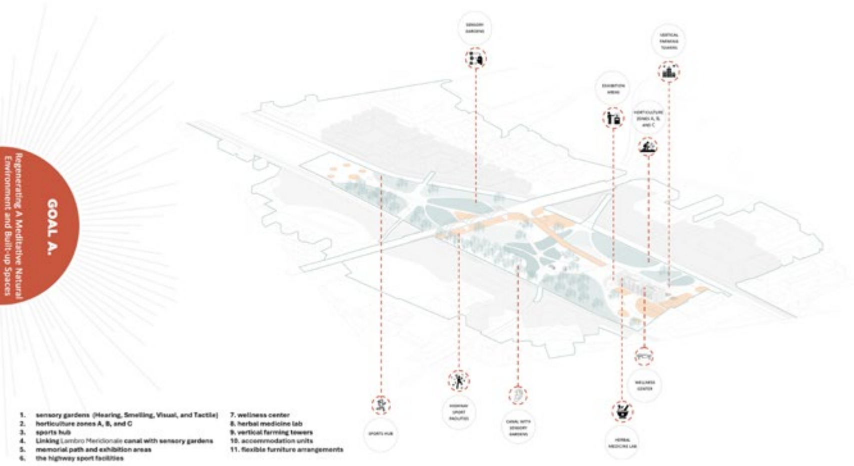
Prioritizing circular economy principles, the masterplan adaptively reuses the site's abandoned industrial buildings to preserve heritage and minimize embodied carbon. These historical structures are retrofitted with energy-efficient envelopes and passive cooling systems, transforming them into highly sustainable community hubs.

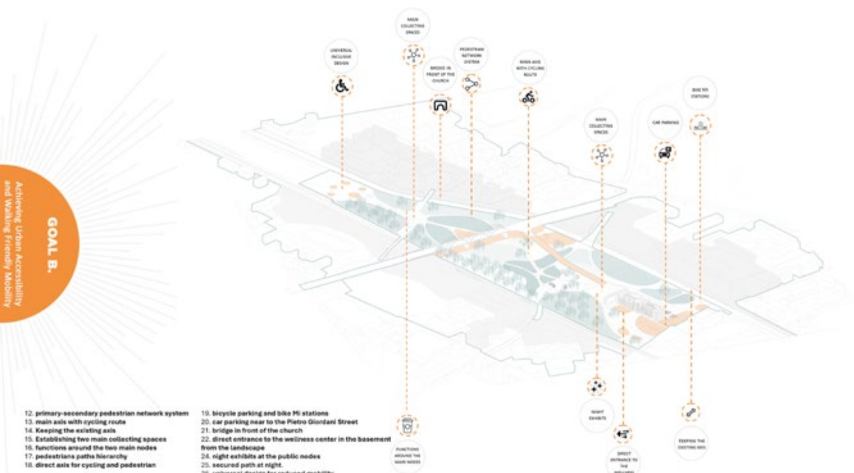


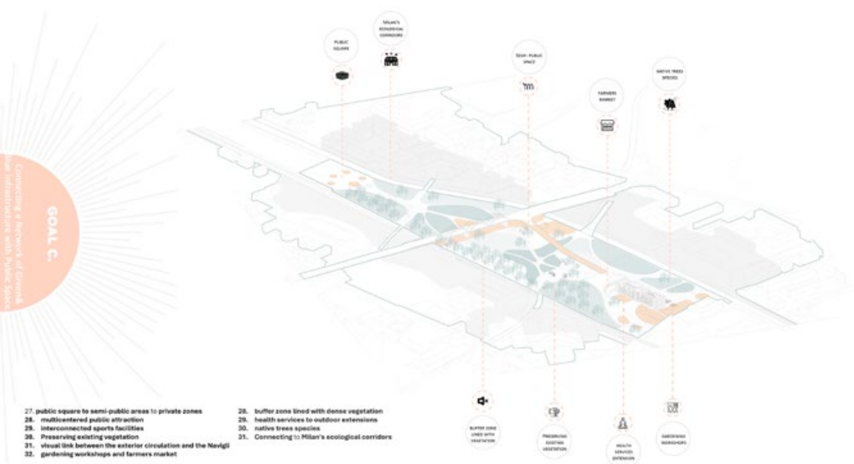
WINTER – RESIDENTIAL SECTION

SUMMER – RESIDENTIAL SECTION

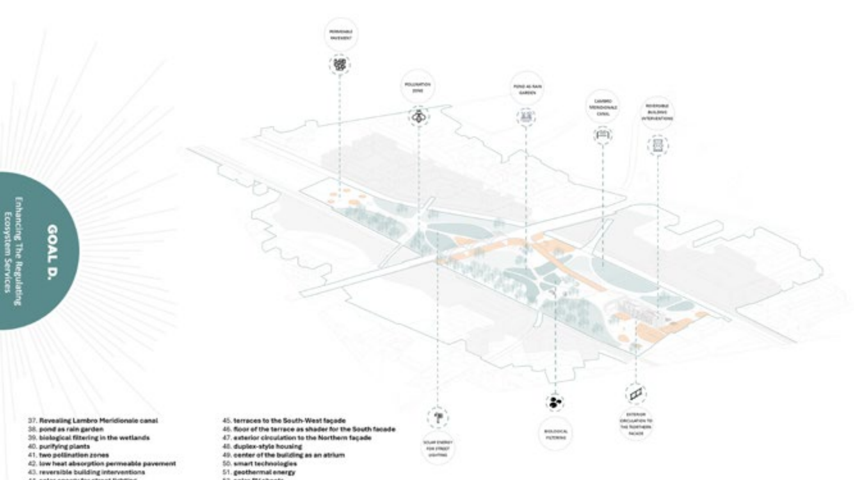
A primary challenge of the site is the severe urban barrier created by the existing railway infrastructure and the Naviglio Grande canal, which historically severed local connectivity. To resolve this, the design introduces a strategic network of pedestrian and cyclist links, creating high-permeability axes that successfully bridge the water and the tracks to stitch the divided urban fabric back together.







Ultimately, the San Cristoforo regeneration goes beyond physical infrastructure and ecological restoration to foster a highly resilient, inclusive community. By integrating flexible public squares, accessible waterfront promenades, and mixed-use programmatic zones, the masterplan establishes a safe, 24-hour destination. This strategy not only encourages healthy social interaction and active urban life but also stimulates local economic growth within the district.



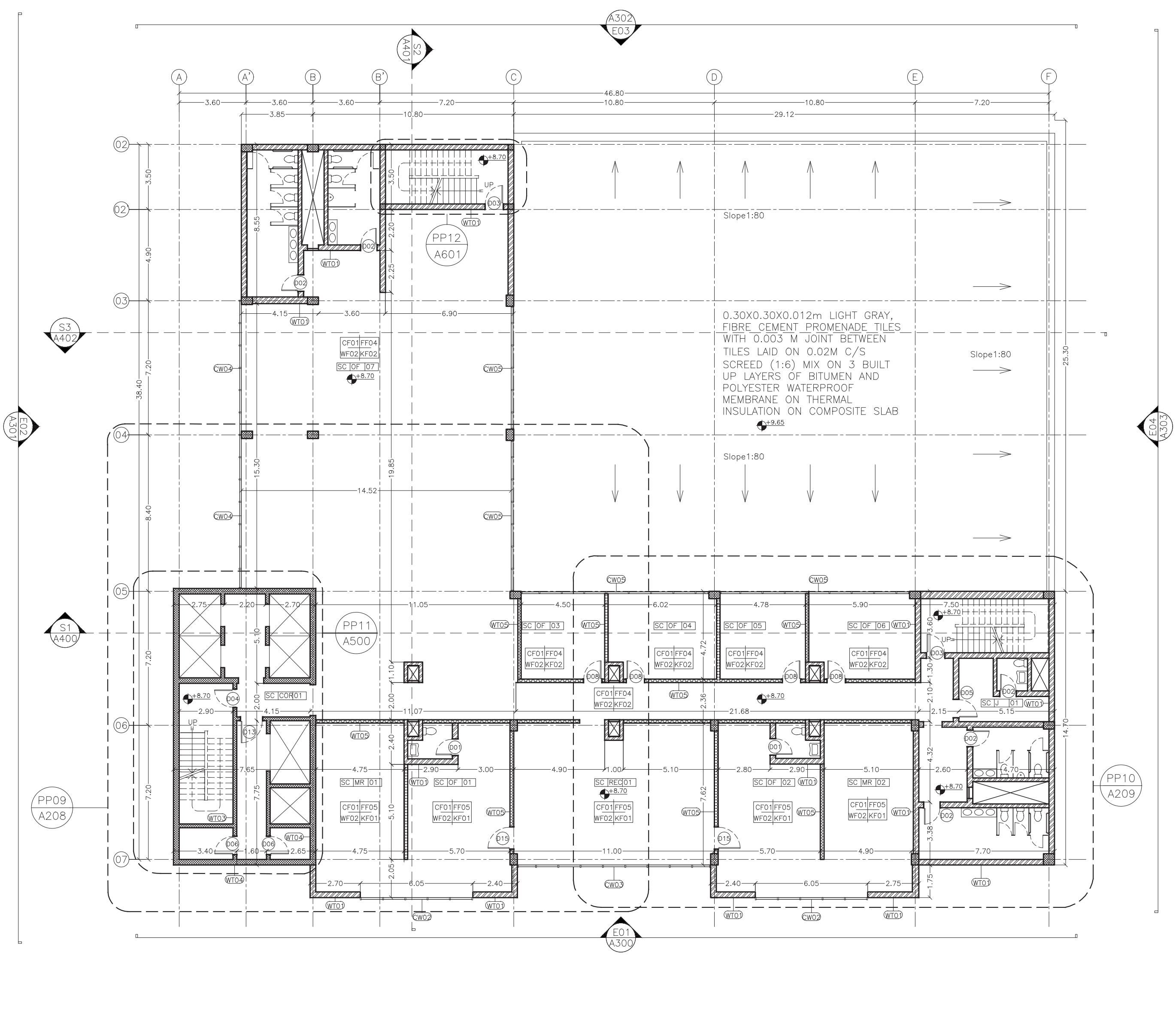
By respectfully preserving the industrial memory of the railway yard while introducing modern, sustainable amenities, the project serves as a forward-thinking model for post-industrial European cities. It successfully connects localized pedestrian networks with Milan's wider metropolitan scale, proving that landscape and infrastructure can seamlessly coexist to transform a historic urban barrier into a vibrant, living gateway.

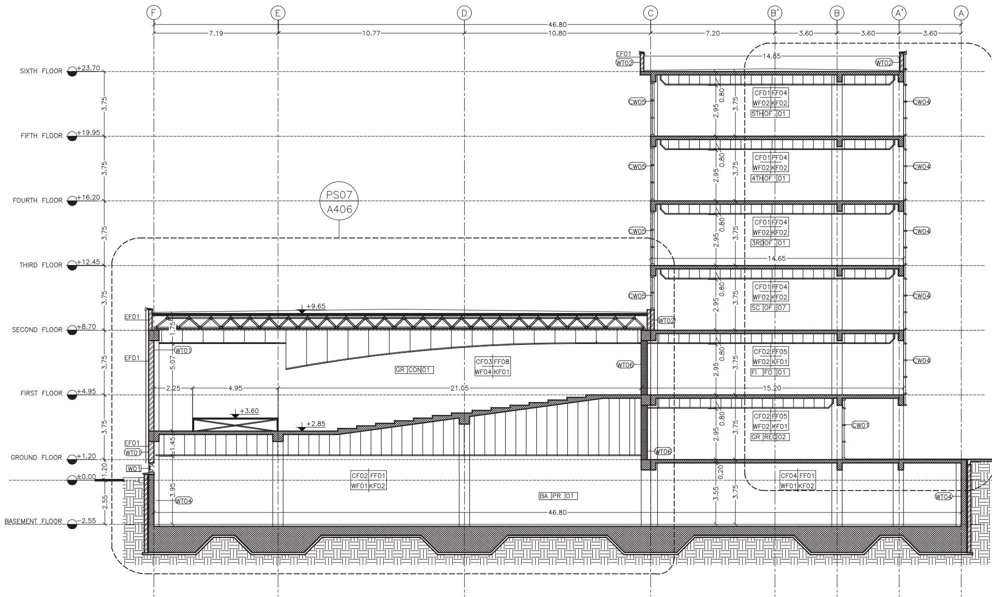
WORKING DRAWING

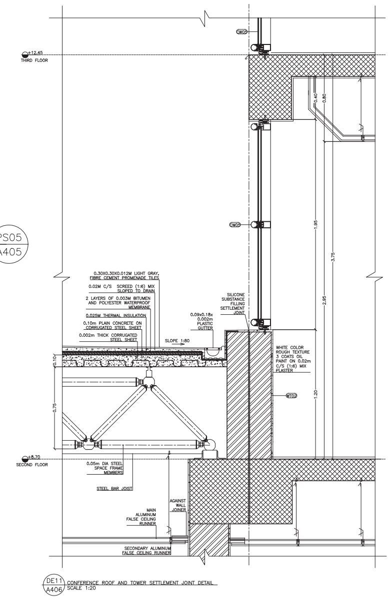
Date and Location:
Oct 2019, Khartoum, Sudan

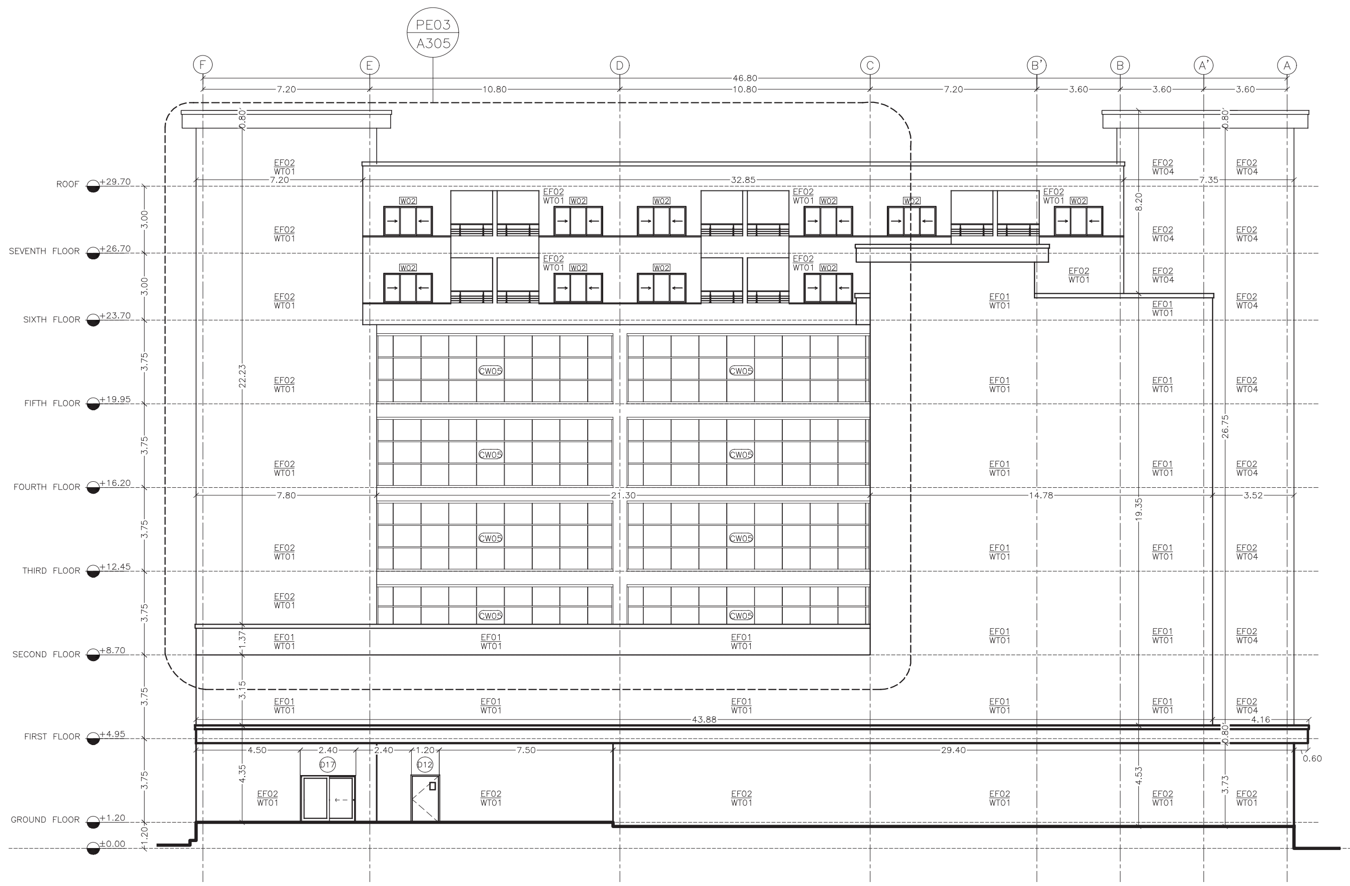
Type:
Academic Studio / Construction Documentation (CD)

Description:
This project translates the conceptual design of a large-scale, mixed-use complex into a comprehensive and rigorously detailed Construction Document (CD) set. To ensure clear communication with future contractors and engineering disciplines, the drafting package strictly adheres to the U.S. National CAD Standard (NCS) and AIA organizational guidelines.









Executed in AutoCAD, the documentation demonstrates advanced technical proficiency. It incorporates rigorous structural grid coordination, detailed material cross-referencing, and complex wall section assemblies, alongside fully resolved vertical circulation cores and schedules. The entire set is systematically managed using professional sheet numbering—categorizing the 200-series for Plans, 300-series for Elevations, and 400-series for Sections—proving strong capability in architectural file management.

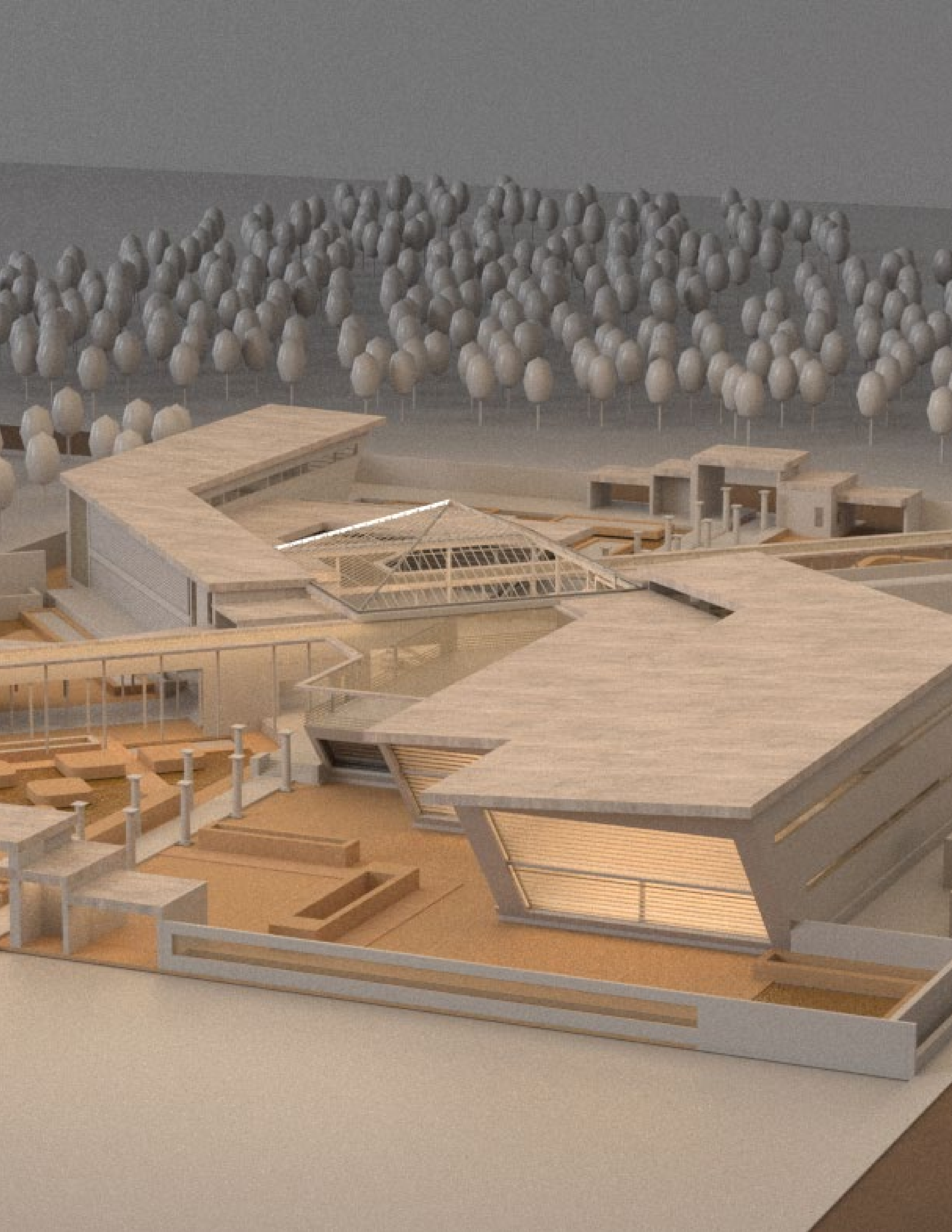
KUSH KINGDOM MUSEUM

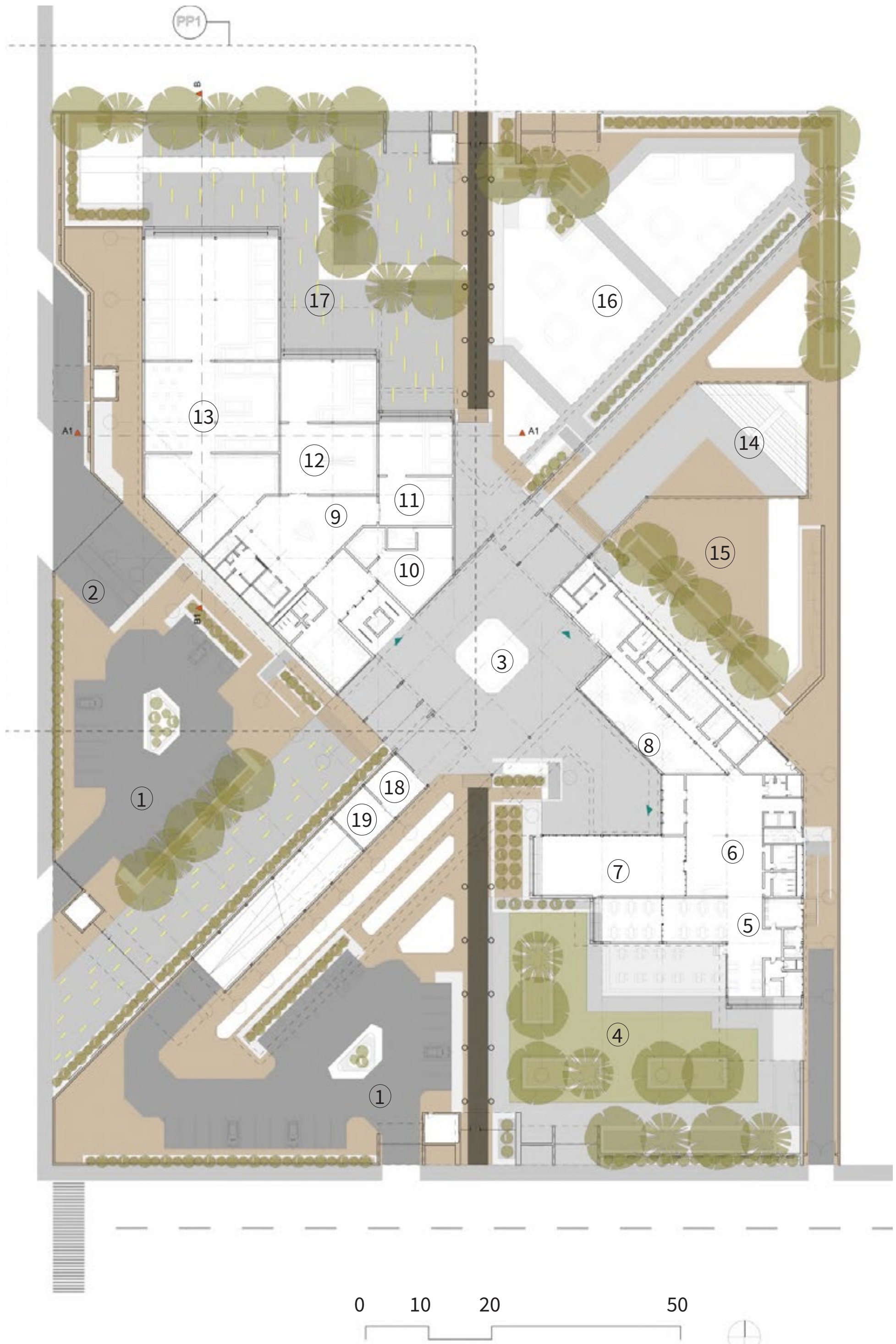
Date and Location:
Oct 2021, Karima, Sudan

Description:
This graduation project addresses a critical crisis in Sudanese archaeology by proposing a modern museum and research center in the historic city of Karima. It replaces degraded storage facilities with a secure, permanent venue to protect and exhibit the irreplaceable artifacts of the Kushite Kingdom, while establishing a dedicated research hub for international excavation missions.

Situated 360 meters from Jebel Barkal's UNESCO World Heritage site, the masterplan directly engages the historic landscape. To protect the fragile archaeological grounds, regional festivals are strategically relocated into a dedicated outdoor amphitheater within the new museum.

The architectural massing is deliberately oriented to frame its sacred context. Elevated exhibition spaces offer sweeping views of Jebel Barkal and its temple ruins, seamlessly linking the indoor artifacts with Kushite spiritual beliefs.





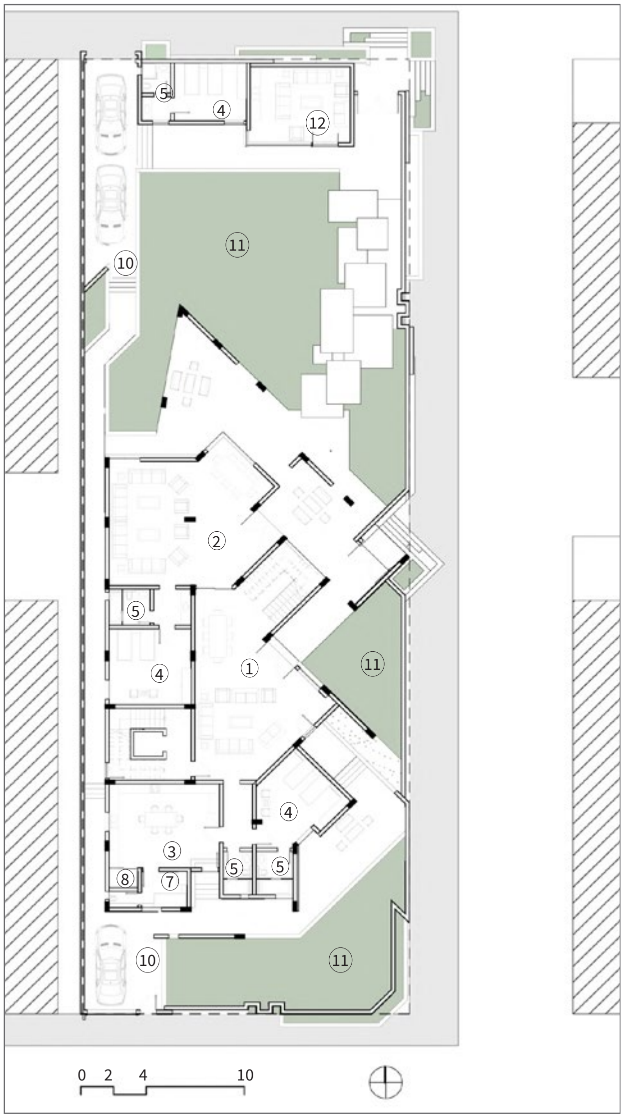
Functionally, the floor plan is divided into two distinct zones to optimize security and operational flow: a controlled curatorial wing for artifact display, and an ancillary zone housing educational and administrative facilities. Within the main galleries, the exhibition is arranged as a chronological journey, guiding visitors through the four major periods of Kushite civilization—from Kerma to Meroe.

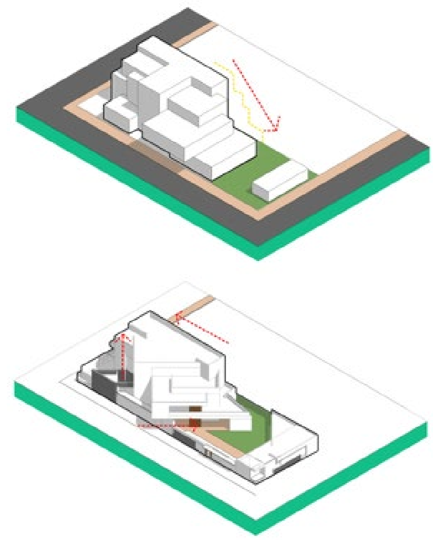
Internally, this spatial layout is strictly governed by the environmental conservation requirements of the artifacts. Visitors move through the sequence organized by material photosensitivity—transitioning from highly climate-controlled, artificially lit galleries for delicate jewelry and fabrics, and culminating in naturally sunlit halls for robust stone monuments.

ALFATIH VILLA

Date and Location:
Jan 2023, Khartoum, Sudan
Type:
Professional Practice (Mo. Siraj Engineering Co. Ltd)

Description:
Designed during my tenure at Mo. Siraj Engineering Co. Ltd, this 1,200 sq.m. residential villa accommodates an extended family of 16. The primary architectural challenge was balancing a luxurious modern aesthetic with strict cultural requirements for absolute internal privacy and gender-separated hosting.





To resolve these programmatic needs, the building mass is stretched longitudinally and rotated on a 45-degree axis, successfully dividing the house into a public-facing northern wing for hosting and a highly private southern block for the family. By stepping the massing—peaking at the rear and degrading towards the street—the design dynamically enhances the visual approach while creating deep, shaded terraces.

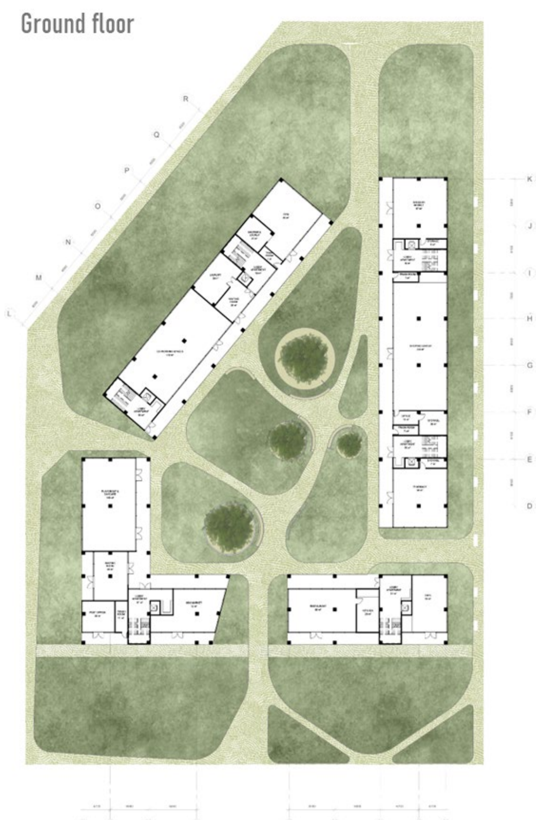
From a technical point of view, the villa utilizes a reinforced concrete frame finished in minimalist grey plaster and wood-textured tiles. A strategic circulation core with four distinct entrances and a central lift ensures independent access to the upper-level apartments while preserving total privacy for the primary residence.

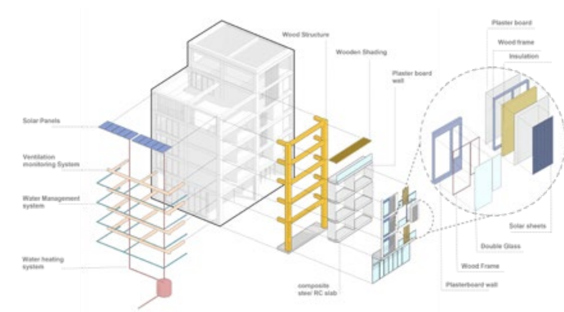
REVIVE BOVISA

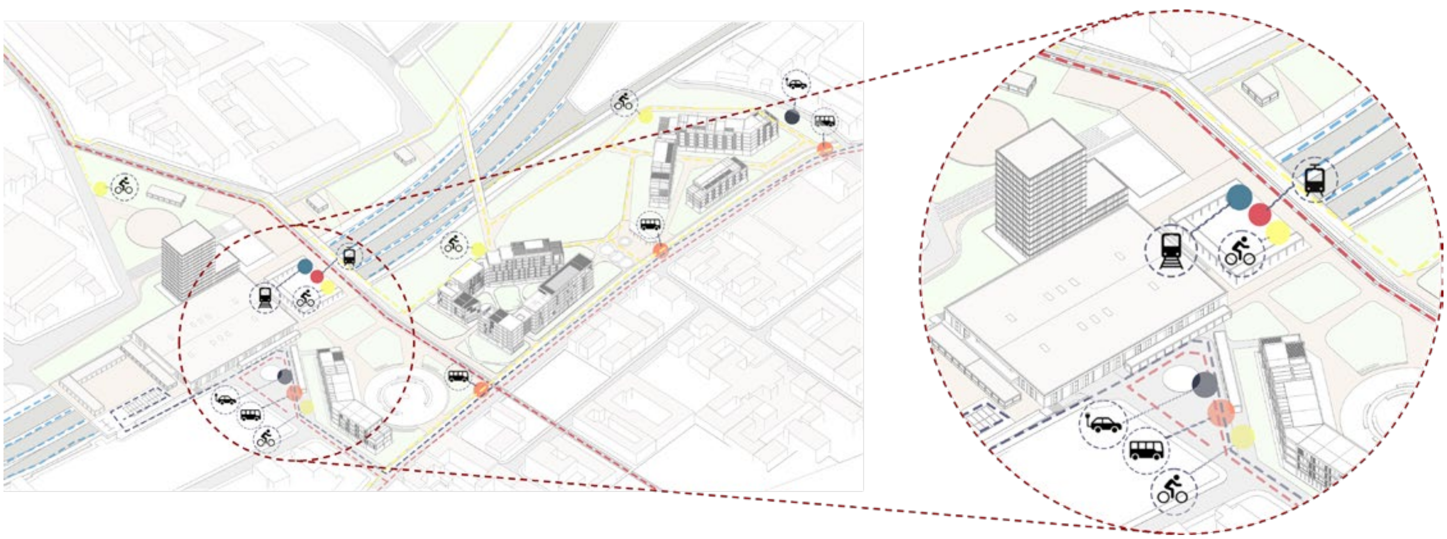
NODO BOVISA REGENERATION

Date and Location:
Sep 2023, Bovisa, Milan, Italy
Type:
Academic Group Project / Urban Design

Description:
Revive Bovisa is a 9-hectare urban regeneration project designed to reconnect a fractured district in Milan into a highly sustainable, intermodal transportation hub.







Developed collaboratively using the Integrated Modification Methodology (IMM), the master plan dedicates 65% of the site to green spaces and active mobility networks. The architectural intervention mirrors Milan's traditional courtyard morphology while integrating mass-timber construction, passive cooling systems, and a diverse functional mix to activate urban metabolism.

To further enhance environmental performance, the buildings incorporate active solar strategies and a central rainwater collection basin designed to naturally cool the microclimate during the summer. Ground-floor co-working and commercial spaces anchor the residential units above, ensuring a vibrant, 24-hour community.

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Architecture speaks loudest in its quietest details—the subtle moments of the built environment that ask us to pause and observe